Justin Yang Chae

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RESEARCH INTERESTS

Reinforcement learning for LLM and VLM reasoning: self-play, RL with verifiable rewards (RLVR), gym-style training environments, and long-horizon agents for training-data curation.

EDUCATION

University of Washington, Seattle Sept 2023 – June 2027 (expected)

B.S. Computer Science and B.A. Mathematics GPA: 3.87/4.0

Relevant coursework: Machine Learning, Deep Learning, Advanced Machine Learning, Natural Language Processing, Computer Vision, Algorithms

PUBLICATIONS

[1] MD Tanvirul Alam, Saksham Aggarwal, **Justin Yang Chae**, Nidhi Rastogi. "SPHINX: A Synthetic Environment for Visual Perception and Reasoning." *CVPR 2026 Findings Workshop*; earlier version at *Multimodal Algorithmic Reasoning (MAR) Workshop, NeurIPS 2025*. [paper]

[2] **Justin Yang Chae**, MD Tanvirul Alam, Nidhi Rastogi. "Towards Understanding Self-play for LLM Reasoning." *5th Workshop on Mathematical Reasoning and AI (MATH-AI), NeurIPS 2025*. [paper]

RESEARCH EXPERIENCE

Prime Intellect June 2026 – Present

RL Resident (Peer: Florian Brand)

- Built PretrainGym, an environment that tests one piece of recursive self-improvement (RSI): a coding agent curates a pretraining corpus for a fixed nanoGPT-style model under a 400M-token budget, rewarded by the trained model's held-out loss minus a decontamination-based leakage penalty. [blog]
- Hardened the environment against reward hacking over three versions after agents inferred the held-out set, first through web search and then through self-scoring; the final version scores on Paloma's 12 sources from a network-free sandbox with an MCP server of five scoped tools.
- Evaluated three frontier coding agents (Kimi K3, GLM-5.3, GPT-6.1 Sol); Kimi K3 scored best (reward 1.042, test loss 3.125) by screening corpora cheaply and tuning mixtures at full budget. Published every run in a turn-by-turn trace viewer.

AI4SEC Lab, Rochester Institute of Technology May 2025 – Aug 2025

REU Intern (Advisors: Prof. Nidhi Rastogi, MD Tanvirul Alam)

- Led a first-author study of asymmetric proposer–solver self-play post-training on 3B and 7B LLMs with verl, comparing it against RLVR and SFT through parameter-update sparsity, token-entropy dynamics, alternative proposer rewards and pass@k [2].
- Showed the performance ceiling of purely on-policy self-play and the case for off-policy support.
- Co-developed SPHINX, a procedurally generated visual perception gym with 25 task types; frontier VLMs such as GPT-4o score 51.1%, and RLVR training on it improves external visual-reasoning benchmarks [1].

Ubiquitous Computing Lab, University of Washington Jan 2025 – Present

Student Researcher (Advisors: Prof. Shwetak Patel, Vidya Srinivas)

- Developed multimodal speaker identification from smart-glasses audio and video streams using audio and image encoders.
- Built an egocentric audio memory system that retrieves conversational snippets for personalized chatbots.
- Engineered an ASR and speaker-diarization pipeline (NVIDIA Parakeet, Pyannote) feeding structured transcripts to LLMs for keyword extraction.
- Prototyped a computer-architecture-inspired memory hierarchy for LLMs, with memory tiers at different lookup latencies; benchmarked related memory modules using Ollama, Docker and LangChain.

HONORS & AWARDS

- Leo Maddox Foundation Endowed Scholarship 2024
- National Merit Scholarship 2023 – 2027
- UW Dean's List, every graded quarter Autumn 2023 – Spring 2026; Annual Dean's List 2023–24, 2024–25

SERVICE

- Reviewer, MATH-AI: The 6th Workshop on Mathematical Reasoning and AI, NeurIPS 2026 — 2026
- Reviewer, MATH-AI: The 5th Workshop on Mathematical Reasoning and AI, NeurIPS 2025 — 2025

SKILLS

Languages: Python, Java, C, JavaScript/TypeScript, SQL, HTML/CSS, LaTeX

ML & research tools: PyTorch, verl, NumPy, Pandas, scikit-learn, Weights & Biases, Matplotlib, Seaborn

Infrastructure: Slurm, Apptainer, Docker, Git, Linux, Cloudflare Tunnels

ACTIVITIES

- Violinist, University of Washington Symphony Orchestra — 2023 – Present
- Violinist, Seattle Youth Symphony Orchestra (1st violin section) — 2014 – 2023